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The Invariance Principle for the Lorentz Mirror Walk in High Dimensions ($d \geq 4$)

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1 Introduction

The *Lorentz mirror walk* is a deterministic dynamical system embedded in a random environment: one may think of a light ray traveling on the integer lattice $\mathbb{Z}^d$ and being deflected by local mirror configurations sampled independently throughout the medium. On large spatial and temporal scales the walk is diffusive, and the purpose of this report is to make this statement precise as a functional central limit theorem, or invariance principle, in the path space $C([0, \Theta], \mathbb{R}^d)$.

The model is a simple way to study *transport in a heterogeneous medium*. A particle in a random medium can display several competing behaviors:

- **kinetic or ballistic transport:** at short times the particle keeps its velocity for long stretches, so its displacement is of order t ;
- **localization:** the particle may become trapped by the geometry of the environment and remain in a bounded region;
- **diffusive transport:** after many changes of direction, the displacement should instead be of order $\sqrt{t}$.

This report focuses on the diffusive phase. The ballistic regime describes the early kinetic behaviour, before enough changes of direction have occurred. Localization would mean that the geometry of the environment traps the trajectory for very long times, or even forever. Diffusion is the intermediate large-scale behaviour in which many microscopic interactions average out: the displacement is no longer linear in time, but has size $\sqrt{t}$, and the rescaled path is expected to converge to Brownian motion.

The Lorentz mirror walk is a good test model for this transition because the randomness is very explicit, while the dynamics still has genuine memory. Here, *memory effects* mean the correlations created by the fixed environment: when the walk comes back to a previously seen site, it sees the same mirror as before rather than a newly sampled one. At long times the trajectory has many opportunities to return near its own past, so the future is influenced by the mirrors discovered earlier. Thus one can study this main difficulty of random transport without the full analytic complexity of models such as the Lorentz gas or random Schrödinger evolutions. The high-dimensional assumption $d \geq 4$ is important in this discussion, because self-intersections are sufficiently rare for these memory effects to remain controllable on the time scales considered here.

1.1 The Lorentz Mirror Walk

We now recall the precise construction of the model used in [1]:

Throughout this subsection, discrete time is indexed by

$$\mathbb{N}_0 := \{0, 1, 2, \dots\}.$$

Let

$$\mathcal{V} := \{\pm e_1, \dots, \pm e_d\}$$

be the set of axis-aligned unit directions in $\mathbb{Z}^d$. A local mirror configuration is described by a map

$$\mu : \mathcal{V} \rightarrow \mathcal{V}$$

belonging to the set

$$\mathcal{M}_d := \left\{ \mu : \mathcal{V} \rightarrow \mathcal{V} \mid \mu(-\mu(v)) = -v \text{ for all } v \in \mathcal{V}, \quad \mu(v) \neq -v \text{ for all } v \in \mathcal{V} \right\}.$$

The condition $\mu(-\mu(v)) = -v$ makes the dynamics time-reversible, while the condition $\mu(v) \neq -v$ excludes head-on collisions. In particular, the identity map belongs to $\mathcal{M}_d$.

The random environment is a field

$$M = (M_x)_{x \in \mathbb{Z}^d}$$

of i.i.d. random variables with values in $\mathcal{M}_d$. For each site $x \in \mathbb{Z}^d$, with probability $1 - p$ we set $M_x = \text{id}$, and with probability p we choose M_x uniformly in $\mathcal{M}_d$. Since $\text{id} \in \mathcal{M}_d$, the total probability that $M_x = \text{id}$ is $1 - p + \frac{p}{|\mathcal{M}_d|}$. The Lorentz mirror walk is the process

$$\mathcal{W}_L(t) = (X_L(t), V_L(t)), \quad t \in \mathbb{N}_0,$$

starting from

$$X_L(0) = 0, \quad V_L(0) = e_1,$$

and evolving according to

$$X_L(t+1) = X_L(t) + V_L(t), \quad V_L(t+1) = M_{X_L(t+1)}(V_L(t)). \quad (1)$$

In words, the particle first moves one step in its current direction, and then the mirror at the arrival site determines its new velocity. Once the environment is fixed, the dynamics are deterministic.

2 Main Result of the Project

2.1 Motivation

Having fixed the formal rules, let us explain why this particular walk is a natural object to study. The walk has just enough structure to capture the essential difficulty of diffusion in a fixed random medium:

- **“fresh” randomness at first visits:** when the particle reaches a new site, it discovers whether a mirror is present there;
- **deterministic memory at revisits:** if the particle returns to an old site, the same mirror acts again, so the process is not Markovian in position and velocity alone;
- **a kinetic scale:** since a non-trivial mirror appears with probability of order p , the typical time between velocity changes is of order p^{-1} .

Thus the model is elementary at the level of its definition, but it still retains the obstruction that makes random transport difficult: the walk repeatedly uses information from the environment it has already discovered. The key question is whether this memory remains weak enough, on time scales much larger than the kinetic scale, for diffusion to emerge.

2.2 Main Result from the Paper

Theorem 2.1. *Let $d \geq 4$. For all p sufficiently small, we have for all times $p^{-5/4} \leq T \leq e^{\log^2(1/p)}$*

$$\left| \frac{p}{T} \mathbb{E}[\|X_L(T)\|_2^2] - \frac{2d-1}{d-1} \right| \leq p^{1/9}.$$

2.3 Statement of the Functional CLT

The functional central limit theorem below is the main statement proved in this report. It should be viewed as a consequence, or byproduct, of the quantitative results of [1] together with a standard invariance-principle argument. The original paper provides the estimates that control the diffusive scale and the effect of self-interactions. Once these inputs are available, the usual Donsker–Prokhorov strategy applies: prove tightness of the rescaled paths and identify the limiting finite-dimensional distributions. The purpose of the later sections is to make this passage explicit.

The regime is dilute in the sense that $p \downarrow 0$, so non-trivial mirrors become rarer in the microscopic environment. At the same time, a fixed macroscopic time $\theta \in [0, \Theta]$ corresponds to $\lfloor \theta T(p) \rfloor$ microscopic steps, and $T(p) \rightarrow \infty$. Thus the walk is observed for longer and longer microscopic times as the environment becomes more dilute. Since the relevant number of potential scattering times is of order $pT(p)$, and $T(p) \gg p^{-11/8}$ implies $pT(p) \rightarrow \infty$, the walk still experiences many effective changes of direction on a fixed macroscopic time interval. The estimates of [1] are precisely what make this dilute long-time regime quantitative; the passage from those estimates to the functional CLT is then the standard part of the argument.

Let $T(p)$ be a microscopic observation time scale satisfying

$$T(p) \gg p^{-11/8}, \quad \Theta T(p) \leq e^{\log^2(1/p)}.$$

This means that the walk is observed far beyond the kinetic scale p^{-1} , while still remaining within the time window controlled by the estimates of [1]. In particular,

$$T(p) \rightarrow \infty \quad \text{as } p \downarrow 0.$$

For example, one may take

$$T(p) = p^{-m}$$

with any fixed $m > 11/8$; for p sufficiently small this satisfies the upper bound above for every fixed Θ .

We first define the rescaled step process associated with the original Lorentz mirror walk:

$$\tilde{Z}_{p,L}(\theta) := \sqrt{\frac{p}{T(p)}} X_L(\lfloor \theta T(p) \rfloor), \quad \theta \in [0, \Theta].$$

This process is a step function, with jumps at the grid points, and is therefore not continuous. To work in the continuous path space $C([0, \Theta], \mathbb{R}^d)$, we use its linear interpolation. Namely, if

$$k = \lfloor \theta T(p) \rfloor,$$

we define

$$Z_{p,L}(\theta) := \sqrt{\frac{p}{T(p)}} [X_L(k) + (\theta T(p) - k)(X_L(k+1) - X_L(k))].$$

Then

$$Z_{p,L} \in C([0, \Theta], \mathbb{R}^d).$$

Moreover, the step process and its continuous interpolation are uniformly asymptotically equivalent. Since the Lorentz mirror walk is nearest-neighbor,

$$\|X_L(k+1) - X_L(k)\| = 1.$$

Therefore,

$$\sup_{\theta \in [0, \Theta]} \|Z_{p,L}(\theta) - \tilde{Z}_{p,L}(\theta)\| \leq \sqrt{\frac{p}{T(p)}}.$$

Since $p \downarrow 0$ and $T(p) \rightarrow \infty$, this upper bound tends to 0. Hence replacing the step process by its continuous interpolation does not change the limiting law.

Theorem 2.2 (Functional CLT for the Lorentz mirror walk). *Let $d \geq 4$. For each sufficiently small $p > 0$, let $Z_{p,L}$ be the linearly interpolated rescaled Lorentz mirror walk defined above, and denote by $\mathbb{P}_p$ its law on $C([0, \Theta], \mathbb{R}^d)$. Then, as $p \downarrow 0$,*

$$Z_{p,L} \Rightarrow B \quad \text{in } C([0, \Theta], \mathbb{R}^d).$$

Here B is a d -dimensional Brownian motion with $\text{Cov}(B(1)) = \sigma^2 I_d$, where

$$\sigma^2 = \frac{2d-1}{d(d-1)}.$$

3 The Lorentz Mirror Walk and the Reference Setting

3.1 Guiding intuition

The idea behind concatenation: A classical Random Walk heuristic says that the amount of intersection between two independent Random Walk depends strongly on the dimension. In dimensions $d \leq 3$, two independent Random Walks have nontrivial intersections. In dimensions $d > 4$, the expected number of intersection opportunities remains uniformly bounded as the observation scale grows. The borderline case is $d = 4$: the expected number of such intersection opportunities diverges, but only logarithmically.

This result is also relevant for our study of the Lorentz mirror walk. A useful way to think about this is to take a Random Walk and cut it at some time t . The part after time t can be viewed as a Random Walk started from the cutting point. By time reversal, the part before time t , seen backwards, can also be viewed as a Random Walk started from the same point. Thus the self-intersections of one Random Walk around time t can be compared heuristically with the intersection of two independent Random Walks started from the same point: one describing the future, and one describing the past.

This idea motivates the concatenation argument. If the walk is already expected to be diffusive on large scales, then its past and future pieces should behave, heuristically, like two Random Walks. Therefore the Random walks intersection picture suggests that self-interactions should be much harder to control in dimensions $d \leq 3$, while in dimensions $d \geq 4$ they can be controlled, with logarithmic losses in the borderline case

$d = 4$.

If interactions with the past are rare once the walk is in the diffusive regime, then a trajectory of length $2t$ should behave like the concatenation of two almost independent trajectories of length t . At the level of the endpoint, this gives the heuristic

$$\frac{X(2t)}{\sqrt{2t/p}} \approx \frac{1}{\sqrt{2}} \left(\frac{X^{(1)}(t)}{\sqrt{t/p}} + \frac{X^{(2)}(t)}{\sqrt{t/p}} \right),$$

where $X^{(1)}$ and $X^{(2)}$ are independent copies. This is the central-limit picture: a centered Gaussian law is a fixed point of the operation

$$Y \mapsto \frac{Y_1 + Y_2}{\sqrt{2}},$$

where Y_1 and Y_2 are independent copies of Y .

The invariance principle is the path-level version of the same idea. If f and g are two independent rescaled paths on $[0, 1]$, then concatenating them and normalizing by $\sqrt{2}$ gives

$$\mathcal{C}(f, g)(\theta) := \begin{cases} 2^{-1/2} f(2\theta), & 0 \leq \theta \leq \frac{1}{2}, \\ 2^{-1/2} (f(1) + g(2\theta - 1)), & \frac{1}{2} \leq \theta \leq 1. \end{cases}$$

Brownian motion is a fixed point of this operation: its two halves are independent increments, and Brownian scaling gives exactly the factor $2^{-1/2}$. This is why the concatenation estimate is the natural path-space analogue of the central-limit renormalization.

How do we control the error coming from the “almost independence” approximation?

The main obstruction is memory. The mirrors are fixed in space. When the walk returns to an old site, it does not see fresh randomness; it sees the same mirror again. Thus the future trajectory is no longer independent of the past. The proof therefore has to show that these memory effects remain small enough to be treated as perturbative errors.

The kinetic scale comes from a simple Bernoulli picture. At each microscopic step, the walk discovers a non-trivial mirror with probability of order p . Thus the lengths of straight segments are approximately geometric random variables with parameter p , and hence have typical size p^{-1} . This is the natural kinetic scale of the model.

On a longer time interval of length t , the walk makes about pt effective changes of direction. Each effective step has length of order p^{-1} , so the expected diffusive displacement is $p^{-1}\sqrt{pt} = \sqrt{\frac{t}{p}}$. This motivates the diffusive rescaling used in the proof. In the proof we often use the enlarged cutoff

$$t_* = \lfloor p^{-1} \log^3(1/p) \rfloor.$$

The remaining difficulty is that this Bernoulli picture is valid only while the walk keeps discovering fresh mirrors. When the walk returns to an old site, it sees the same mirror again, creating memory effects. This memory error cannot be controlled as a single global defect: it is the cumulative effect of many local opportunities for self-interaction. The proof therefore controls it step by step, showing at each concatenation scale that the probability of hitting the past remains small.

3.2 The Kesten–Papanicolaou comparison

The first technical step is the short-time comparison of Kesten and Papanicolaou. Before the walk hits its previously discovered environment, the old mirrors are irrelevant. On that event, the Lorentz mirror walk evolves exactly like the driving process of Definition 3.1: the particle usually keeps its velocity, and, when it does turn, it chooses a new non-opposite direction according to the mirror law.

For the comparison one uses a slightly enlarged version of the kinetic scale,

$$t_* := \lfloor p^{-1} \log^3(1/p) \rfloor$$

where the logarithmic factor absorbs the errors in the short-time estimates.

At time t , let $M(t)$ denote the mirrors discovered since the last regeneration time $\alpha(t)$ in the notation of [1]; to avoid confusion with the environment field $M = (M_x)_{x \in \mathbb{Z}^d}$, we denote this set by $\mathcal{D}(t)$ here:

$$\mathcal{D}(t) := \{0\} \cup \{X_{\text{reg}}(s) : s \in \mathcal{T} \cap [\alpha(t), t]\}.$$

The reference paper then introduces the predictable set of dangerous times

$$S := \left\{ s > 0 : \begin{array}{l} \exists y \in \mathcal{D}(s-1), \quad \exists u \neq \pm V(s-1), \\ \exists r \in [0, 2t_*] \text{ such that } y + ru = X_{\text{reg}}(s) \end{array} \right\}.$$

Thus $s \in S$ means that, with one suitable turn, the walk could quickly point toward an old mirror. If no actual mirror discovery time lies in S , and if no straight flight is abnormally long, then no relevant self-interaction occurs. This gives the short-time coupling estimate

$$\mathbb{P}(X_{\text{reg}}(t) = \tilde{X}(t) \text{ for all } t \leq T) \geq 1 - p^3 T^2 \log^6(1/p)$$

in the range where the Kesten–Papanicolaou argument is applied. The same section computes the variance of the driving process and transfers this short-time diffusive variance to the regenerated walk.

Thus the Kesten–Papanicolaou comparison gives the memory-free base case of the proof: as long as the walk does not interact with its past, it can be replaced by the Markovian driving process. The later argument keeps this picture as the reference model and shows that the errors caused by occasional memory effects can be treated as small perturbations.

3.3 The Driving Process

The next constructions extend this comparison to the regime where memory effects are present but rare. The *driving process* is the clean Markovian walk obtained by always using fresh randomness. The driven walk is coupled to it and deviates only when the already discovered environment forces a correction. In this way, the proof starts from the Kesten–Papanicolaou base model and then controls the true mirror walk as a perturbation of the driving walk.

The paper [1] also introduces two auxiliary constructions that play a central role in the proof and that we shall use repeatedly below: the *driving process* and the *regenerated walk*. The estimates in the reference are organized through a long-time induction. This means that one first proves the relevant bounds at a short, essentially kinetic scale,

and then proves an induction step: assuming the transition and traveling-far estimates hold up to a time T , the concatenation argument propagates them to a slightly longer time. Iterating this step gives control up to the maximal time horizon $\bar{T}$. The driving process is the Markovian model with which the mirror walk can be compared before it sees its own past. The regenerated walk is the object on which this long-time induction is carried out.

Definition 3.1 (Driving process). *Fix initial data*

$$\tilde{X}(0) \in \mathbb{Z}^d, \quad \tilde{V}(0) \in \mathcal{V}.$$

Let $\mathcal{T} \subset \mathbb{N}_0$ be a random subset obtained by keeping each time $t \in \mathbb{N}_0$ independently with probability p . These are the times at which a new mirror is discovered.

For $t \notin \mathcal{T}$, set $\tilde{\mu}(t) = \text{id}$, and for $t \in \mathcal{T}$ let $\tilde{\mu}(t)$ be an independent uniform element of $\mathcal{M}_d$. Then define, for $t \geq 1$,

$$\tilde{X}(t) = \tilde{X}(t-1) + \tilde{V}(t-1), \quad \tilde{V}(t) = \tilde{\mu}(t)[\tilde{V}(t-1)].$$

The position process $\tilde{X}$ is therefore a non-backtracking walk on $\mathbb{Z}^d$.

3.4 The Driven Process

Definition 3.2 (Construction of the driven walk from the driving process). *Fix initial conditions*

$$X(0) \in \mathbb{Z}^d, \quad V(0) \in \mathcal{V}.$$

Given a driving process

$$\tilde{W}(t) := (\tilde{X}(t), \tilde{V}(t), \mathbf{1}_{\{t \in \mathcal{T}\}}, \tilde{\mu}(t)),$$

we define a driven walk

$$\Phi(\tilde{W}) = (X, V, \mu)$$

recursively as follows. Set $\mu(0) = \tilde{\mu}(0)$. For $t \geq 1$, define

$$X(t) = X(t-1) + V(t-1), \quad V(t) = \mu(t)[V(t-1)],$$

where $\mu(t)$ is chosen according to the following rules:

- (1) If $X(t) = X(s)$ for some $s \in [0, t)$, then we set $\mu(t) = \mu(s)$.
- (2) If $X(t) \neq X(s)$ for all $s \in [0, t)$ and $V(t-1) = \tilde{V}(t-1)$, then we set $\mu(t) = \tilde{\mu}(t)$.
- (3) If $X(t) \neq X(s)$ for all $s \in [0, t)$, $V(t-1) \neq \tilde{V}(t-1)$, and $\tilde{\mu}(t)[\tilde{V}(t-1)] = -V(t-1)$, then we also set

$$\mu(t) = \tilde{\mu}(t).$$

- (4) Otherwise, if $X(t) \neq X(s)$ for all $s \in [0, t)$,

$$V(t-1) \neq \tilde{V}(t-1), \quad \tilde{\mu}(t)[\tilde{V}(t-1)] \neq -V(t-1),$$

then write

$$a := V(t-1), \quad b := \tilde{V}(t-1), \quad A := \tilde{\mu}(t)[a], \quad B := \tilde{\mu}(t)[b].$$

We define $\mu(t)$ by swapping the two outgoing directions, together with the corresponding time-reversed entries:

$$\mu(t)[a] = B, \quad \mu(t)[b] = A, \quad \mu(t)[-B] = -a, \quad \mu(t)[-A] = -b,$$

and

$$\mu(t)[v] = \tilde{\mu}(t)[v]$$

for every

$$v \notin \{a, b, -A, -B\}.$$

The condition in this case prevents a head-on collision, and the two assignments on $-A$ and $-B$ preserve the reversibility relation defining $\mathcal{M}_d$.

This is the coupling used in [1]; it is designed so that the driven walk follows the driving walk whenever this is compatible with the already discovered environment.

The four cases in Definition 3.2 describe how the driven walk is coupled to the driving walk while remaining compatible with the environment already discovered by the trajectory. There are two basic situations:

- **old site:** if the walk returns to a previously discovered site, it must reuse the mirror already assigned to that site;
- **fresh site:** if the site has not been visited before, the walk tries to use the mirror proposed by the driving process.

The only genuinely corrective step is rule (4). In that case the site is fresh, but the driven and driving velocities do not match. The proposed mirror is then locally modified so that the driven walk can realign with the driving walk after the interaction, while still satisfying the reversibility constraint defining $\mathcal{M}_d$.



Figure 1: Two basic ways in which the driven walk can fail to follow the driving process directly. In case (a), the walk arrives at a site where a mirror has already been discovered, so it must reuse the old mirror. In case (b), the site is fresh, but the mirror proposed by the driving process may not be compatible with the current coupling between the driven and driving trajectories.

In the notation of rule (4), the local modification is

$$\tilde{\mu}: \quad a \mapsto A, \quad b \mapsto B, \quad \mu: \quad a \mapsto B, \quad b \mapsto A, \quad -B \mapsto -a, \quad -A \mapsto -b.$$

Thus the correction is local: only the entries needed for the coupling, together with their time-reversed partners, are changed.



Figure 2: Illustration of coupling rule (4). In panel (a), the driving walk encounters a mirror. In panel (b), the driven walk encounters a modified mirror chosen so that, after the interaction, the two walks leave the site with matching outgoing directions. This is the mechanism used to keep the two trajectories coupled while respecting the previously discovered environment.

3.5 The Regenerated Mirror Walk and Why It Is Used

Definition 3.3 (Regenerated mirror walk). *Let*

$$\{(X_L^i(s), V_L^i(s))\}_{i \geq 1}$$

be i.i.d. Lorentz mirror walks, each started from the origin with initial direction e_1 , and let

$$\tau_{\text{clo}}^i := \min\{t > 0 : X_L^i(t) = 0 \text{ and } V_L^i(t) = e_1\}$$

be the closing time of the i -th walk. Define the regeneration times by

$$\tau_{\text{reg}}^0 := 0, \quad \tau_{\text{reg}}^i := \sum_{j=1}^i \tau_{\text{clo}}^j, \quad i \geq 1.$$

The regenerated walk

$$\mathcal{W}_{\text{reg}}(s) = (X_{\text{reg}}(s), V_{\text{reg}}(s))$$

is obtained by concatenating these independent closed pieces:

$$X_{\text{reg}}(s) = X_L^i(s - \tau_{\text{reg}}^{i-1}), \quad V_{\text{reg}}(s) = V_L^i(s - \tau_{\text{reg}}^{i-1}), \quad s \in [\tau_{\text{reg}}^{i-1}, \tau_{\text{reg}}^i]. \quad (2)$$

In words, the walk follows one mirror trajectory until it closes, then restarts from a fresh independent copy. We write

$$\alpha(t)$$

for the last regeneration time before t . When the initial velocity is chosen uniformly in $\mathcal{V}$, we denote the corresponding regenerated walk by

$$\mathcal{W}_{\text{reg}}^* = (X_{\text{reg}}^*, V_{\text{reg}}^*).$$

The point of this construction is the following. If the original Lorentz walk returns to the origin with the same velocity, then, because the environment is fixed, it is forced to repeat the same loop. The regenerated walk cuts the trajectory at exactly this closing time and restarts with an independent environment. Thus $\mathcal{W}_{\text{reg}}$ agrees pathwise with the original Lorentz walk $\mathcal{W}_L$ up to the first closing time τ_{clo} , but after such a closing time it produces a fresh block rather than repeating a deterministic cycle. This makes $\mathcal{W}_{\text{reg}}$ the right object for the multiscale argument: each regenerated block has the correct local law, and different blocks can be used as independent pieces in the concatenation estimates.

Since the traveling-far estimate $H2$, which gives a tail estimate on the regenerated

walk, and the concatenation arguments used later are formulated for the regenerated walk, we write $X_{\text{reg}}(t)$ for its position process. The original Lorentz walk will always be denoted by $X_L(t)$. At the very end of the proof in [1], estimates for $\mathcal{W}_{\text{reg}}$ are transferred back to the original walk $\mathcal{W}_L$ by showing that early closing is rare on the time window under consideration.

The original mirror walk is not Markovian in its position and velocity alone. When it returns to an old site, the mirror is the same one as before, and the future depends on the explored environment. The regenerated walk removes only the most rigid obstruction: when the path closes by returning to the origin with the same velocity, the original walk would repeat the same loop forever, so the regenerated walk starts a fresh independent copy instead.

This is why the induction in [1] is carried out for $\mathcal{W}_{\text{reg}}$, not directly for $\mathcal{W}_L$. Before the first closing time the two processes coincide exactly. After a regeneration, one should not expect pathwise closeness to the original walk anymore; the purpose is instead to produce fresh blocks with the correct law. These blocks are the raw material for the long-time concatenation argument.

3.6 Induction Hypotheses

It is useful to keep the main quantitative targets visible. In [1], the central time horizon is

$$\bar{T} := \left\lfloor e^{\log^2(1/p)} \right\rfloor.$$

The induction hypothesis statements are:

- **Induction hypothesis (H1): transition probabilities.** Fix $\varepsilon := 0.01$. For any $z \in \mathbb{Z}^d$,

$$\mathbb{P}(\|X_{\text{reg}}(T) - z\| \leq p^{-1}) \leq (pT)^{-d/2} p^{-\varepsilon} e^{(\log T)^{1/3}}.$$

- **Induction hypothesis (H2): traveling far.** For any $s \leq t \leq T$,

$$\mathbb{P}\left(\|X_{\text{reg}}(t) - X_{\text{reg}}(s)\| \geq \log^8(1/p) \sqrt{\frac{t-s}{p}}\right) \leq e^{-2\log^2(1/p)}.$$

- **Notation.** We write $H1(T)$ when (H1) holds at time T , and $H2(T)$ when (H2) holds up to time T . We write $H(T)$ when both hypotheses hold for all times $0 \leq t \leq T$.

4 Ideas Behind the Diffusivity Estimate

4.1 Self-interactions and the local tests

In Section 5 of [1], the point is to detect, before attempting a concatenation, whether the current environment contains too much of the past near the walk.

The main local objects are the following:

- **Too many discoveries in a kinetic window:**

$$\tau_{\text{many}} := \min \left\{ t \geq p^{-1} : |\mathcal{T} \cap [t - p^{-1}, t]| \geq \log^3(1/p) \right\}.$$

- **Too long a straight flight:**

$$\tau_{\text{few}} := \min \{t \geq t_* : V(t) = V(t') \text{ for all } t' \in [t - t_*, t]\}.$$

- **First future interaction with the past:** $\tau_{\text{hit}}(t)$ denotes the first time after t at which the future walk interacts with the relevant past of the current regenerated block. Up to this time, the future can be coupled to a fresh regenerated walk started from $(X_{\text{reg}}(t), V_{\text{reg}}(t))$.

The global obstruction is measured by heavy blocks. For $r \geq p^{-1}$, a block $B(x, r)$ is called **heavy** at time t if

$$|B(x, r) \cap \mathcal{D}(t)| \geq p^{1.9} r^2.$$

Such a block is dangerous because, even if no short window contains too many newly discovered mirrors, repeated moderate visits can make one spatial region dense enough to force future self-interactions.

The local obstruction is measured by considering the set of relaxed times. A time t is **relaxed** if:

- (i) for every $r \geq p^{-1}$, the block $B(X_{\text{reg}}(t), r)$ is not heavy at time t ;
- (ii) for every $1 \leq s \leq t_*$,

$$X_{\text{reg}}(t) + sV_{\text{reg}}(t) \notin \mathcal{D}(t).$$

The first condition removes the large-scale obstruction, while the second removes the ballistic obstruction: the walk is not immediately heading along a straight segment into an old mirror.

The corresponding bad stopping times are

$$\tau_{\text{hea}} := \min \{t > 0 : \text{there exists a heavy block at time } t\},$$

$$\tau_{\text{rel}} := \min \{t > t_* : |R \cap [t - t_*, t]| \leq 0.9t_*\}, \quad \tau := \tau_{\text{hea}} \wedge \tau_{\text{rel}} \wedge \tau_{\text{many}} \wedge \tau_{\text{few}},$$

where R is the set of relaxed times.

Under $H(T)$, the key estimates of Section 5 of [1] are:

- **escape from a relaxed time:**

$$\mathbb{P}(\tau_{\text{hit}}(t) \geq (t + T) \wedge \tau \mid \mathcal{F}_t) \geq \frac{3}{4} \quad \text{on } \{t \text{ is relaxed}\};$$

- **bad geometry is unlikely:**

$$\mathbb{P}(\tau \leq T) \leq e^{-c \log^3(1/p)}.$$

These estimates are exactly what is needed to make the concatenation in Section 6 work.

4.2 Concatenation and induction

We define the concatenation

$$\mathcal{W}_c = (X_c, V_c)$$

of

$$\mathcal{W}_{\text{reg},1}^* = (X_{\text{reg},1}^*, V_{\text{reg},1}^*) \quad \text{and} \quad \mathcal{W}_{\text{reg},2}^* = (X_{\text{reg},2}^*, V_{\text{reg},2}^*)$$

by

$$X_c(s) := \begin{cases} X_{\text{reg},1}^*(s), & s \in [0, t_1], \\ X_{\text{reg},1}^*(t_1) + X_{\text{reg},2}^*(s - t_1), & s \in [t_1, t_1 + t_2]. \end{cases} \quad (3)$$

Our goal is to show that the concatenation $\mathcal{W}_c$ can be coupled with the regenerated mirror walk $\mathcal{W}_{\text{reg}} = (X_{\text{reg}}, V_{\text{reg}})$ of length $t_1 + t_2$, starting from direction e_1 .

The proof of such a coupling goes as follows. Assuming $H(T)$, the coupling repeatedly waits for times at which both walks are relaxed. At such a common relaxed time, each future has a fixed positive probability of avoiding its own past; on that event the two walks can be driven by the same future randomness and become synchronized again. Bounding the number and length of the failed synchronization intervals gives Proposition 6.1:

Proposition 4.1 (Proposition 6.1 of [1]). *Assume that $H(T)$ holds for some $T \leq \bar{T}$, and let $t_1, t_2 \geq 0$ satisfy $t_1 + t_2 \leq T + 1$. Let $\mathcal{W}_{\text{reg},1}^*$ and $\mathcal{W}_{\text{reg},2}^*$ be two independent regenerated mirror walks with uniform initial velocity, and let $\mathcal{W}_c = (X_c, V_c)$ be their concatenation at time t_1 . Let $\mathcal{W}_{\text{reg}} = (X_{\text{reg}}, V_{\text{reg}})$ be a regenerated mirror walk of length $t_1 + t_2$ with initial velocity e_1 . Then there exists a coupling of $\mathcal{W}_c$ and $\mathcal{W}_{\text{reg}}$ such that*

$$\mathbb{P}\left(\max_{t \leq t_1 + t_2} \|X_{\text{reg}}(t) - X_c(t)\| \geq p^{-1} \log^7(1/p)\right) \leq e^{-c \log^3(1/p)}.$$

The same conclusion holds if $\mathcal{W}_{\text{reg}}$ is replaced by a regenerated mirror walk $\mathcal{W}_{\text{reg}}^*$ with uniform initial velocity.

The missing final step is Section 7 of [1], where this concatenation estimate is used to close the induction. The logic is:

$$H(T) \implies \text{control of self-interactions} \implies \text{concatenation} \implies H(T + 1).$$

More concretely, Section 7 has three roles:

- **Variance control.** For the regenerated walk with uniform initial velocity, if σ_t^2 denotes one diagonal entry of the covariance matrix, Lemma 7.1 proves

$$c \frac{t}{p} \leq \sigma_t^2 \leq C \frac{t}{p}, \quad p^{-1} \leq t \leq T,$$

and, for $p^{-5/4} \leq t \leq T$,

$$\left| \frac{p}{t} \sigma_t^2 - \frac{2d-1}{d(d-1)} \right| \leq Cp^{1/8} \log^{15}(1/p).$$

- **Propagation of H_2 .** One decomposes an increment $X_{\text{reg}}(t) - X_{\text{reg}}(s)$ into shorter blocks, replaces the walk by a concatenated walk, and applies concentration to the resulting independent block increments.

- **Propagation of $H1$.** One again replaces the long walk by many independent regenerated blocks, normalizes their variance using Lemma 7.1, and applies anti-concentration for sums of independent variables. The coupling error from Proposition 6.1 is small compared with the kinetic ball of radius p^{-1} .

This proves $H(T+1)$ and therefore, by induction, $H(\bar{T})$. Theorem 3.2 then gives the diffusive second-moment estimate for the regenerated walk stated in Subsection 3.6. Finally, [1] transfers the result from the regenerated walk back to the original Lorentz walk. Since $X_L(t) = X_{\text{reg}}(t)$ on the event $\{\tau_{\text{clo}} > t\}$, the difference between the two processes is supported on the rare early-closing event. Corollary 3.3 of [1] gives

$$\mathbb{P}(\tau_{\text{clo}} \leq \bar{T}) \leq p^{1/3},$$

and the tail estimate controls the size of the displacement on this exceptional event:

$$\frac{p}{T} |\mathbb{E}[\|X_{\text{reg}}(T)\|^2] - \mathbb{E}[\|X_L(T)\|^2]| \leq \frac{1}{2}p^{1/9}.$$

Combining this with Theorem 3.2 gives Theorem 1.1 of [1]. Thus the regenerated walk is not just a technical convenience: it is the object on which the proof can be run cleanly, and the rare-closing estimate is what connects it back to the original Lorentz mirror walk.

5 Strategy of the Proof and Reduction to the Regenerated Walk

The purpose of this section is to explain how the proof of the functional CLT is organized. The technical estimates needed for tightness and for convergence of finite-dimensional distributions are proved in Sections 6 and 7. Here we isolate the general probabilistic structure of the argument.

The proof follows the classical two-step scheme for weak convergence in path space:

- (i) prove *tightness* of the rescaled laws in $C([0, \Theta], \mathbb{R}^d)$;
- (ii) identify every possible limit by proving convergence of finite-dimensional distributions.

The first step gives compactness of the family of laws, through Prokhorov's theorem. The second step identifies the only possible limit as Brownian motion with diffusivity

$$\sigma^2 = \frac{2d-1}{d(d-1)}.$$

Thus the proof is reduced to two concrete tasks: tightness of the paths and convergence of finite-dimensional distributions.

The main input behind this scheme is the collection of estimates proved in [1]: the second-moment estimate, the traveling-far estimate $H2$, and the concatenation coupling. The previous sections explained why the driving process, the regenerated walk, relaxed times, and concatenation are the right objects to use. In the rest of the report we use these estimates as inputs and show how they imply the functional CLT.

5.1 Reduction to the Regenerated Walk

We first prove the functional CLT for the regenerated walk. Recall that X_{reg} denotes the regenerated Lorentz mirror walk. Define the rescaled step process

$$\tilde{Z}_{p,\text{reg}}(\theta) := \sqrt{\frac{p}{T(p)}} X_{\text{reg}}(\lfloor \theta T(p) \rfloor), \quad \theta \in [0, \Theta].$$

Since this process is a step function, we work in the continuous path space $C([0, \Theta], \mathbb{R}^d)$ by taking its linear interpolation. Namely, for

$$k = \lfloor \theta T(p) \rfloor,$$

set

$$Z_{p,\text{reg}}(\theta) := \sqrt{\frac{p}{T(p)}} [X_{\text{reg}}(k) + (\theta T(p) - k)(X_{\text{reg}}(k+1) - X_{\text{reg}}(k))].$$

Then $Z_{p,\text{reg}} \in C([0, \Theta], \mathbb{R}^d)$ almost surely.

The original Lorentz mirror walk and the regenerated walk coincide up to the first closing time τ_{clo} . Since

$$\Theta T(p) \leq \bar{T} := \left\lfloor e^{\log^2(1/p)} \right\rfloor$$

for p sufficiently small, Corollary 3.3 of [1] gives

$$\mathbb{P}(\tau_{\text{clo}} \leq \Theta T(p)) \leq \mathbb{P}(\tau_{\text{clo}} \leq \bar{T}) \leq p^{1/3}.$$

Consequently,

$$\mathbb{P}(Z_{p,L} = Z_{p,\text{reg}} \text{ on } [0, \Theta]) \longrightarrow 1.$$

Therefore it is enough to prove the functional CLT for $Z_{p,\text{reg}}$. The convergence for $Z_{p,L}$ will then follow by Slutsky's theorem.

We denote by σ^2 the scalar diffusivity constant appearing in the limiting Gaussian law.

5.2 Continuity and Interpolation

We first record two elementary facts about the interpolation. They justify working in the continuous path space $C([0, \Theta], \mathbb{R}^d)$, and show that the interpolated process and the step process have the same weak limits.

Lemma 5.1 (Continuity of the interpolated process). *For any fixed $p \in (0, 1)$, the linearly interpolated process $Z_{p,\text{reg}}$ belongs almost surely to the continuous path space $C([0, \Theta], \mathbb{R}^d)$.*

Proof. For each integer $k \geq 0$, the function $\theta \mapsto Z_{p,\text{reg}}(\theta)$ is affine on the interval

$$[k/T(p), (k+1)/T(p)],$$

hence continuous there. At the grid points $\theta = k/T(p)$, the left and right affine pieces both take the value

$$Z_{p,\text{reg}}\left(\frac{k}{T(p)}\right) = \sqrt{\frac{p}{T(p)}} X_{\text{reg}}(k),$$

so the pieces match continuously. Therefore $Z_{p,\text{reg}} \in C([0, \Theta], \mathbb{R}^d)$ almost surely. $\square$

Lemma 5.2 (Uniform asymptotic equivalence). *As $p \rightarrow 0$,*

$$\sup_{\theta \in [0, \Theta]} \|Z_{p, \text{reg}}(\theta) - \tilde{Z}_{p, \text{reg}}(\theta)\| \rightarrow 0$$

deterministically.

Proof. For $k = \lfloor \theta T(p) \rfloor$,

$$Z_{p, \text{reg}}(\theta) - \tilde{Z}_{p, \text{reg}}(\theta) = \sqrt{\frac{p}{T(p)}} (\theta T(p) - k) (X_{\text{reg}}(k+1) - X_{\text{reg}}(k)).$$

Since the fractional part is at most 1 and the walk is nearest-neighbor,

$$\|X_{\text{reg}}(k+1) - X_{\text{reg}}(k)\| = 1,$$

we get

$$\sup_{\theta \in [0, \Theta]} \|Z_{p, \text{reg}}(\theta) - \tilde{Z}_{p, \text{reg}}(\theta)\| \leq \sqrt{\frac{p}{T(p)}}.$$

Since $T(p) \rightarrow \infty$, the right-hand side tends to zero. This proves the claim. $\square$

5.3 The Two Technical Inputs

Sections 6 and 7 prove the two ingredients needed for weak convergence.

First, Section 6 proves tightness of the laws of $Z_{p, \text{reg}}$. The main idea is to compare the regenerated walk with a concatenation of independent regenerated blocks. The block skeleton records only the endpoints of these blocks. Its increments are independent and centered, so fourth-moment estimates can be used to control its modulus of continuity. The oscillations inside the blocks are then shown to be negligible using the traveling-far estimate $H2$.

The tightness argument only requires

$$T(p) \gg p^{-5/4}.$$

The stronger assumption

$$T(p) \gg p^{-11/8}$$

is imposed in Theorem 2.2 because it is needed later in the proof of finite-dimensional convergence.

Second, Section 7 proves convergence of finite-dimensional distributions. Unlike tightness, which requires uniform control of the whole path, the FDD argument only concerns finitely many increments

$$Z_{p, \text{reg}}(\theta_i) - Z_{p, \text{reg}}(\theta_{i-1}).$$

The concatenation coupling reduces these increments to sums of independent regenerated blocks. The variance estimate from [1] identifies the limiting covariance, while Lindeberg's theorem gives Gaussianity and independence of the limiting increments.

5.4 Conclusion from Tightness and Finite-Dimensional Distributions

We now explain why these two ingredients imply the main theorem. The following proposition is the final step: once tightness and finite-dimensional convergence have been proved for the regenerated walk, the desired functional CLT for the original Lorentz walk follows.

Proposition 5.3 (Conclusion of the proof). *Assume that:*

- (i) *the family of laws of $Z_{p,\text{reg}}$ is tight in $C([0, \Theta], \mathbb{R}^d)$;*
- (ii) *for every $k \geq 1$ and every $0 \leq \theta_1 < \dots < \theta_k \leq \Theta$,*

$$(Z_{p,\text{reg}}(\theta_1), \dots, Z_{p,\text{reg}}(\theta_k)) \xrightarrow{d} (B(\theta_1), \dots, B(\theta_k)),$$

where B is a d -dimensional Brownian motion with $\text{Cov}(B(1)) = \sigma^2 I_d$.

Then

$$Z_{p,L} \Rightarrow B \quad \text{in } C([0, \Theta], \mathbb{R}^d).$$

Proof. By tightness and Prokhorov's theorem, the laws of $Z_{p,\text{reg}}$ are relatively compact in the weak topology on $C([0, \Theta], \mathbb{R}^d)$. Hence every sequence $p_n \downarrow 0$ has a subsequence $p_{n_j} \downarrow 0$ such that the laws of $Z_{p_{n_j},\text{reg}}$ converge weakly to some probability measure $\mathbb{P}^*$ on $C([0, \Theta], \mathbb{R}^d)$.

We identify $\mathbb{P}^*$. Since the evaluation maps

$$f \mapsto (f(\theta_1), \dots, f(\theta_k))$$

are continuous on $C([0, \Theta], \mathbb{R}^d)$, weak convergence in $C([0, \Theta], \mathbb{R}^d)$ implies convergence of finite-dimensional distributions. Therefore the finite-dimensional distributions of $\mathbb{P}^*$ coincide with the limits of the finite-dimensional distributions of $Z_{p,\text{reg}}$. By assumption, these are the finite-dimensional distributions of Brownian motion with diffusivity σ^2 .

Probability measures on $C([0, \Theta], \mathbb{R}^d)$ are uniquely determined by their finite-dimensional distributions; see, for instance, [2], Theorem 12.3. Hence $\mathbb{P}^*$ must be the Wiener measure associated with B . Thus every possible limit is the same Wiener measure, so the whole family $Z_{p,\text{reg}}$ converges weakly to B :

$$Z_{p,\text{reg}} \Rightarrow B \quad \text{in } C([0, \Theta], \mathbb{R}^d).$$

Finally, by the rare-closing estimate,

$$\mathbb{P}(Z_{p,L} = Z_{p,\text{reg}} \text{ on } [0, \Theta]) \longrightarrow 1.$$

Equivalently,

$$\|Z_{p,L} - Z_{p,\text{reg}}\|_\infty \xrightarrow{\mathbb{P}} 0.$$

By Slutsky's theorem,

$$Z_{p,L} \Rightarrow B \quad \text{in } C([0, \Theta], \mathbb{R}^d).$$

□

6 Tightness

Remark 6.1 (Choice of the time scale). *In this tightness proof we restrict, for simplicity of notation, to polynomial time scales*

$$T(p) = p^{-m}, \quad m > \frac{5}{4}.$$

This is not essential for the strategy. The argument only needs a number of blocks $n_p \rightarrow \infty$ such that, with

$$r_p := \frac{T(p)}{n_p},$$

the following two conditions hold:

$$r_p \geq p^{-5/4}$$

so that each block lies in the range where the estimates of [1] are available, and

$$\sqrt{\frac{p}{T(p)}} n_p p^{-1} \log^7(1/p) \rightarrow 0$$

so that the concatenation coupling error vanishes after rescaling. Equivalently, one needs

$$1 \ll n_p \ll \min \left\{ T(p)p^{5/4}, \frac{\sqrt{pT(p)}}{\log^7(1/p)} \right\}.$$

Thus the same proof applies to more general time scales satisfying

$$T(p) \gg p^{-5/4}, \quad \Theta T(p) \leq e^{\log^2(1/p)}.$$

The stronger lower bound $T(p) \gg p^{-11/8}$ is only used later for FDD convergence. For clarity, we present the proof in the polynomial case $T(p) = p^{-m}$, where one can simply take

$$n_p = \lfloor p^{-\beta} \rfloor, \quad 0 < \beta < \min \left\{ m - \frac{5}{4}, \frac{m-1}{2} \right\}.$$

Proof structure.

- Construct a global concatenated walk $Z_{p,c}$ by splitting $[0, \Theta T(p)]$ into N_p independent regenerated blocks of length $r_p \simeq T(p)/n_p$.
- Introduce the block skeleton $\widehat{Z}_{p,c}$, which records only the endpoints of the independent blocks and interpolates linearly between them.
- Prove a modulus-of-continuity estimate for $\widehat{Z}_{p,c}$ using independence of the blocks, fourth-moment bounds, Markov's inequality, and a union bound.
- Show that internal block oscillations are negligible:

$$\left\| Z_{p,c} - \widehat{Z}_{p,c} \right\|_{\infty} \xrightarrow[p \downarrow 0]{\mathbb{P}} 0.$$

- Use the iterated concatenation coupling to transfer the modulus estimate from $Z_{p,c}$ to $Z_{p,\text{reg}}$:

$$\|Z_{p,\text{reg}} - Z_{p,c}\|_\infty \xrightarrow[p \downarrow 0]{\mathbb{P}} 0.$$

- Conclude that $Z_{p,\text{reg}}$ satisfies the standard modulus-of-continuity criterion for tightness in $C([0, \Theta], \mathbb{R}^d)$.

We prove tightness of the regenerated process $Z_{p,\text{reg}}$ in $C([0, \Theta], \mathbb{R}^d)$. This is the compactness input needed later to combine finite-dimensional convergence with convergence in path space.

Throughout this section, let

$$T(p) = p^{-m}, \quad m > \frac{5}{4},$$

and set $L := \log(1/p)$. Choose

$$0 < \beta < \min \left\{ m - \frac{5}{4}, \frac{m-1}{2} \right\}.$$

Define

$$n_p := \lfloor p^{-\beta} \rfloor, \quad r_p := \left\lfloor \frac{T(p)}{n_p} \right\rfloor.$$

Thus $n_p \rightarrow \infty$, while

$$r_p \simeq \frac{T(p)}{n_p} = p^{-m+\beta} \rightarrow \infty.$$

The condition $\beta < m - 5/4$ gives $r_p \geq p^{-5/4}$ for p small, so each block lies in the time range where Lemma 7.1 of [1] applies. Moreover, since $T(p) = p^{-m} = e^{mL}$ and $\bar{T} = e^{L^2}$, we have $\Theta T(p) \leq \bar{T}$ for p small.

We now construct a concatenated walk on $[0, \Theta T(p)]$. Let

$$0 = s_0 < s_1 < \cdots < s_{N_p}$$

be block endpoints such that

$$s_j - s_{j-1} \in \{r_p, r_p + 1\}, \quad s_{N_p} \geq \Theta T(p), \quad N_p \leq C_\Theta n_p.$$

For each j , let $X_{\text{reg}}^{(j),*}$ be an independent regenerated walk with uniform initial velocity, and define the displacement of the j -th block by

$$\Delta_j := X_{\text{reg}}^{(j),*}(s_j - s_{j-1}).$$

We also write for the rescaled increments:

$$\xi_j := \sqrt{\frac{p}{T(p)}} \Delta_j, \quad j = 1, \dots, N_p.$$

The concatenated walk X_c is obtained by placing these independent blocks one after another.

Define

$$Z_{p,c}(\theta) := \sqrt{\frac{p}{T(p)}} X_c(\lfloor \theta T(p) \rfloor), \quad \theta \in [0, \Theta],$$

with linear interpolation between microscopic times. We also define the block skeleton

$$\widehat{Z}_{p,c} \left(\frac{s_k}{T(p)} \right) := \sqrt{\frac{p}{T(p)}} \sum_{j=1}^k \Delta_j, \quad k = 0, \dots, N_p,$$

and extend it linearly between consecutive block endpoints.

For $f \in C([0, \Theta], \mathbb{R}^d)$, write

$$\omega_f(\delta) := \sup_{\substack{0 \leq \theta, \eta \leq \Theta \\ |\theta - \eta| \leq \delta}} \|f(\theta) - f(\eta)\|.$$

By Theorem A.10, since $Z_{p,\text{reg}}(0) = 0$, it is enough to prove that, for every $\rho > 0$,

$$\lim_{\delta \downarrow 0} \limsup_{p \downarrow 0} \mathbb{P}(\omega_{Z_{p,\text{reg}}}(\delta) > \rho) = 0.$$

Lemma 6.2 (Local oscillation of the block skeleton). *Let $I \subset [0, \Theta]$ be an interval of length at most 3δ . Then*

$$\mathbb{E} \left[\sup_{\theta, \eta \in I} \left\| \widehat{Z}_{p,c}(\theta) - \widehat{Z}_{p,c}(\eta) \right\|^4 \right] \leq C \left[\left(\delta + \frac{1}{n_p} \right)^2 + \left(\delta + \frac{1}{n_p} \right) \frac{L^{32}}{n_p} + \varepsilon_p \right],$$

where $\varepsilon_p \rightarrow 0$.

Proof. A block has rescaled length approximately $r_p/T(p) \simeq 1/n_p$. Hence an interval I of length at most 3δ intersects at most

$$k \leq C(\delta n_p + 1)$$

blocks.

Let $q, \dots, q+k-1$ be the block indices whose time intervals intersect I , where q is the first such block index and k is the number of intersected blocks, and define

$$M_\ell := \sum_{i=q}^{q+\ell-1} \xi_i, \quad 0 \leq \ell \leq k.$$

After subtracting the value of $\widehat{Z}_{p,c}$ at the block endpoint immediately before I , the oscillation on I is controlled by the oscillation of these partial sums. The linear interpolation between block endpoints cannot create a larger oscillation than the endpoints. Therefore

$$\sup_{\theta, \eta \in I} \left\| \widehat{Z}_{p,c}(\theta) - \widehat{Z}_{p,c}(\eta) \right\| \leq 2 \max_{0 \leq \ell \leq k} \|M_\ell\|.$$

By Lemma A.8 and Lemma A.9,

$$\mathbb{E} \left[\sup_{\theta, \eta \in I} \left\| \widehat{Z}_{p,c}(\theta) - \widehat{Z}_{p,c}(\eta) \right\|^4 \right] \leq C \left[\left(\frac{k}{n_p} \right)^2 + \frac{kL^{32}}{n_p^2} + k\varepsilon'_p \right].$$

Using $k \leq C(\delta n_p + 1)$, we have

$$\frac{k}{n_p} \leq C \left(\delta + \frac{1}{n_p} \right), \quad \frac{kL^{32}}{n_p^2} \leq C \left(\delta + \frac{1}{n_p} \right) \frac{L^{32}}{n_p}.$$

Finally, since $n_p \varepsilon'_p \rightarrow 0$, the term $k\varepsilon'_p$ is bounded by a new sequence $\varepsilon_p \rightarrow 0$. This proves the estimate. $\square$

Lemma 6.3 (Modulus estimate for the block skeleton). *For every $\rho > 0$,*

$$\lim_{\delta \downarrow 0} \limsup_{p \downarrow 0} \mathbb{P} \left(\omega_{\widehat{Z}_{p,c}}(\delta) > \rho \right) = 0.$$

Proof. We cover $[0, \Theta]$ by enlarged intervals

$$I_\ell := [\ell\delta, (\ell+3)\delta] \cap [0, \Theta], \quad \ell = 0, 1, \dots, \left\lceil \frac{\Theta}{\delta} \right\rceil.$$

The length 3δ is only used to avoid boundary issues: if $|\theta - \eta| \leq \delta$, then θ and η belong to at least one of these intervals.

Set

$$A_\ell := \left\{ \sup_{\theta, \eta \in I_\ell} \left\| \widehat{Z}_{p,c}(\theta) - \widehat{Z}_{p,c}(\eta) \right\| > \rho \right\}.$$

Then

$$\{\omega_{\widehat{Z}_{p,c}}(\delta) > \rho\} \subset \bigcup_{\ell \leq C_\Theta/\delta} A_\ell.$$

By the union bound and Markov's inequality,

$$\mathbb{P} \left(\omega_{\widehat{Z}_{p,c}}(\delta) > \rho \right) \leq \sum_{\ell \leq C_\Theta/\delta} \mathbb{P}(A_\ell) \leq \frac{C_\Theta}{\delta \rho^4} \left[\left(\delta + \frac{1}{n_p} \right)^2 + \left(\delta + \frac{1}{n_p} \right) \frac{L^{32}}{n_p} + \varepsilon_p \right].$$

For fixed $\delta > 0$, $n_p \rightarrow \infty$, $L^{32}/n_p \rightarrow 0$, and $\varepsilon_p \rightarrow 0$. Hence

$$\limsup_{p \downarrow 0} \mathbb{P} \left(\omega_{\widehat{Z}_{p,c}}(\delta) > \rho \right) \leq \frac{C_\Theta}{\rho^4} \delta.$$

Letting $\delta \downarrow 0$ proves the claim. □

Lemma 6.4 (Internal block oscillations). *We have*

$$\left\| Z_{p,c} - \widehat{Z}_{p,c} \right\|_\infty \xrightarrow[p \downarrow 0]{\mathbb{P}} 0.$$

Proof. We need to control the oscillation of X_c inside each block. If $a < b$ are in the same block and $b - a < p^{-1}$, then the nearest-neighbor property gives

$$\|X_c(b) - X_c(a)\| \leq b - a < p^{-1}.$$

Since $r_p \geq p^{-5/4}$, the threshold $CL^8 \sqrt{r_p/p}$ is larger than p^{-1} for p small, so the desired bound is deterministic in this case.

It remains to consider pairs with $b - a \geq p^{-1}$. For such pairs, *H2* applies. Since $b - a \leq 2r_p$, after increasing C ,

$$\mathbb{P} \left(\|X_c(b) - X_c(a)\| \geq CL^8 \sqrt{\frac{r_p}{p}} \right) \leq Ce^{-2L^2}.$$

We now take a union bound over all microscopic pairs inside the blocks. Each block contains at most Cr_p^2 pairs (a, b) , and there are $N_p \leq C_\Theta n_p$ blocks. Since $r_p \simeq T(p)/n_p$, the total number of pairs is at most

$$CN_p r_p^2 \leq C_\Theta n_p \left(\frac{T(p)}{n_p} \right)^2 \leq C_\Theta T(p)^2.$$

Therefore

$$\mathbb{P} \left(\max_{\substack{a < b \\ a, b \text{ in the same block}}} \|X_c(b) - X_c(a)\| \geq CL^8 \sqrt{\frac{r_p}{p}} \right) \leq C_\Theta T(p)^2 e^{-2L^2}.$$

Since $T(p) = p^{-m} = e^{mL}$, the right-hand side is $C_\Theta e^{2mL-2L^2} \rightarrow 0$.

After rescaling, the internal oscillation threshold becomes

$$\sqrt{\frac{p}{T(p)}} CL^8 \sqrt{\frac{r_p}{p}} = CL^8 \sqrt{\frac{r_p}{T(p)}} \leq \frac{CL^8}{\sqrt{n_p}} \rightarrow 0.$$

Thus the maximal rescaled oscillation of $Z_{p,c}$ inside a block vanishes in probability.

Since $\widehat{Z}_{p,c}$ linearly connects the block endpoints,

$$\left\| Z_{p,c} - \widehat{Z}_{p,c} \right\|_\infty$$

is bounded by this maximal internal oscillation. The claim follows. $\square$

Proposition 6.5 (Modulus estimate for the concatenated process). *For every $\rho > 0$,*

$$\lim_{\delta \downarrow 0} \limsup_{p \downarrow 0} \mathbb{P}(\omega_{Z_{p,c}}(\delta) > 2\rho) = 0.$$

Proof. For every $\delta > 0$,

$$\omega_{Z_{p,c}}(\delta) \leq \omega_{\widehat{Z}_{p,c}}(\delta) + 2 \left\| Z_{p,c} - \widehat{Z}_{p,c} \right\|_\infty.$$

Therefore

$$\mathbb{P}(\omega_{Z_{p,c}}(\delta) > 2\rho) \leq \mathbb{P}(\omega_{\widehat{Z}_{p,c}}(\delta) > \rho) + \mathbb{P}\left(2 \left\| Z_{p,c} - \widehat{Z}_{p,c} \right\|_\infty > \rho\right).$$

The second term vanishes in $\limsup_{p \downarrow 0}$ by Lemma 6.4, and the first term is controlled by Lemma 6.3. The claim follows. $\square$

Lemma 6.6 (Coupling transfer). *There exists a coupling of $Z_{p,\text{reg}}$ and $Z_{p,c}$ such that*

$$\|Z_{p,\text{reg}} - Z_{p,c}\|_\infty \xrightarrow[p \downarrow 0]{\mathbb{P}} 0.$$

Proof. By the iterated concatenation coupling from Proposition 6.1 / Corollary 6.2 of [1], applied to the $N_p \leq C_\Theta n_p$ blocks, there exists a coupling such that

$$\mathbb{P} \left(\max_{0 \leq t \leq \Theta T(p)} \|X_{\text{reg}}(t) - X_c(t)\| \geq C_\Theta n_p p^{-1} L^7 \right) \leq C_\Theta n_p e^{-cL^3}.$$

The right-hand side tends to zero, since n_p is polynomial in p^{-1} .

After multiplying by $\sqrt{p/T(p)}$, the coupling error becomes

$$\sqrt{\frac{p}{T(p)}} n_p p^{-1} L^7 = p^{\frac{m-1}{2} - \beta + o(1)} L^7.$$

This tends to zero because $\beta < (m-1)/2$. Hence

$$\|Z_{p,\text{reg}} - Z_{p,c}\|_\infty \xrightarrow[p \downarrow 0]{\mathbb{P}} 0.$$

$\square$

Theorem 6.7 (Tightness of the regenerated process). *The family of laws of $Z_{p,\text{reg}}$ is tight in*

$$C([0, \Theta], \mathbb{R}^d).$$

Proof. Since $Z_{p,\text{reg}}(0) = 0$, it remains to prove the modulus-of-continuity estimate. For every $\delta > 0$,

$$\omega_{Z_{p,\text{reg}}}(\delta) \leq \omega_{Z_{p,c}}(\delta) + 2 \|Z_{p,\text{reg}} - Z_{p,c}\|_\infty.$$

Thus, for every $\rho > 0$,

$$\begin{aligned} \mathbb{P}(\omega_{Z_{p,\text{reg}}}(\delta) > 3\rho) &\leq \mathbb{P}(\omega_{Z_{p,c}}(\delta) > 2\rho) \\ &\quad + \mathbb{P}(2 \|Z_{p,\text{reg}} - Z_{p,c}\|_\infty > \rho). \end{aligned}$$

Taking $\limsup_{p \downarrow 0}$, the second term vanishes by Lemma 6.6, while the first term is controlled by Proposition 6.5. Therefore

$$\lim_{\delta \downarrow 0} \limsup_{p \downarrow 0} \mathbb{P}(\omega_{Z_{p,\text{reg}}}(\delta) > 3\rho) = 0.$$

This is the standard modulus-of-continuity criterion for tightness in $C([0, \Theta], \mathbb{R}^d)$. Hence the laws of $Z_{p,\text{reg}}$ are tight. $\square$

7 Convergence of Finite-Dimensional Distributions

Theorem 7.1 (FDD Convergence). *For every $k \geq 1$ and every finite collection of macroscopic times*

$$0 \leq \theta_1 < \theta_2 < \dots < \theta_k \leq \Theta,$$

the joint distribution of

$$(Z_{p,\text{reg}}(\theta_1), \dots, Z_{p,\text{reg}}(\theta_k))$$

converges weakly, as $p \rightarrow 0$, to that of

$$(B(\theta_1), \dots, B(\theta_k)),$$

where B is a d -dimensional Brownian motion satisfying $\text{Cov}(B(t)) = t\sigma^2 I_d$.

Proof. By Lemma 5.2, it is enough to prove the convergence for the step process $\tilde{Z}_{p,\text{reg}}$. If $\theta_1 = 0$, then the first coordinate is deterministic and equal to 0, both for $\tilde{Z}_{p,\text{reg}}$ and for the limiting Brownian motion. We may therefore omit this coordinate and assume, without loss of generality, that $0 < \theta_1 < \dots < \theta_k$.

Step 1: Define the rescaled grid and the partition.

Fix $0 \leq \theta_1 < \dots < \theta_k \leq \Theta$ and set $\theta_0 := 0$. Define

$$t_i := \lfloor \theta_i T(p) \rfloor, \quad m_i := t_i - t_{i-1}, \quad i = 1, \dots, k.$$

Then

$$\Delta_p^{(i)} := \tilde{Z}_{p,\text{reg}}(\theta_i) - \tilde{Z}_{p,\text{reg}}(\theta_{i-1}) = \sqrt{\frac{p}{T(p)}} (X_{\text{reg}}(t_i) - X_{\text{reg}}(t_{i-1})).$$

For each i , let

$$n_i := \lfloor (m_i p)^{1/3} \rfloor, \quad t'_i := \left\lfloor \frac{m_i}{n_i} \right\rfloor, \quad r_i := m_i - n_i t'_i.$$

Here n_i is the number of subintervals, t'_i is the regular block length, and r_i is the remainder. Since $m_i \sim (\theta_i - \theta_{i-1})T(p)$, we have

$$n_i \sim ((\theta_i - \theta_{i-1})pT(p))^{1/3} \longrightarrow \infty,$$

and

$$t'_i \sim \frac{m_i}{n_i} \sim (\theta_i - \theta_{i-1})^{2/3} T(p)^{2/3} p^{-1/3}.$$

At this point we want the regular block length to satisfy

$$t'_i \geq p^{-5/4},$$

because this is the regime where the precise diffusive variance estimate from [1] applies. Since

$$t'_i \sim (\theta_i - \theta_{i-1})^{2/3} T(p)^{2/3} p^{-1/3},$$

this condition is implied by

$$T(p)^{2/3} p^{-1/3} \gg p^{-5/4}.$$

Equivalently,

$$T(p)^{2/3} \gg p^{-11/12},$$

that is,

$$T(p) \gg p^{-11/8}.$$

Thus, under the standing assumption $T(p) \gg p^{-11/8}$, we indeed have $t'_i \geq p^{-5/4}$ for p sufficiently small.

In the polynomial case $T(p) = p^{-m}$, we have

$$t'_i \sim (\theta_i - \theta_{i-1})^{2/3} p^{-(2m+1)/3}.$$

Therefore $t'_i \geq p^{-5/4}$ for p small provided

$$\frac{2m+1}{3} > \frac{5}{4},$$

that is,

$$m > \frac{11}{8}.$$

Also $0 \leq r_i < t'_i$, and all these times are still much smaller than $\bar{T}$, since $\Theta T(p) \leq \bar{T}$ for p sufficiently small.

Step 2: Construct the global independent concatenated system.

For each macroscopic increment $i = 1, \dots, k$, let

$$X_1^{*,i}, \dots, X_{n_i}^{*,i}, X_{n_i+1}^{*,i}$$

be independent regenerated mirror walks, each started from the origin with a uniform random initial velocity. The first n_i walks have length t'_i , and the last one has length r_i . All these walks are independent as i and j vary.

Define, for each i , the local concatenation

$$X^{*,i}(s)$$

on $[0, m_i]$ by concatenating

$$X_1^{*,i}, \dots, X_{n_i}^{*,i}, X_{n_i+1}^{*,i}$$

on the subgrid

$$0, t'_i, 2t'_i, \dots, n_i t'_i, m_i.$$

Thus

$$X^{*,i}(m_i) = \sum_{j=1}^{n_i} X_j^{*,i}(t'_i) + X_{n_i+1}^{*,i}(r_i).$$

Now define the global concatenated walk X_c on $[0, t_k]$ by placing these local concatenations one after another:

$$X_c(s) := \sum_{\ell=1}^{i-1} X^{*,\ell}(m_\ell) + X^{*,i}(s - t_{i-1}), \quad s \in (t_{i-1}, t_i].$$

Equivalently, X_c is a single concatenation of

$$N := \sum_{i=1}^k (n_i + 1)$$

independent regenerated mirror walks.

We apply Corollary 6.2 of [1] once, globally, to the whole partition of $[0, t_k]$ induced by the subgrids above. The total time satisfies

$$t_k \leq \Theta T(p) \leq \bar{T}$$

for p sufficiently small. Moreover, the regular block lengths satisfy $t'_i \gg p^{-5/4}$, as shown in Step 1, and the possible remainder lengths r_i are allowed in Corollary 6.2. Therefore there exists a coupling between the regenerated walk X_{reg} on $[0, t_k]$ and the global concatenation X_c such that

$$\mathbb{P} \left(\max_{0 \leq s \leq t_k} \|X_{\text{reg}}(s) - X_c(s)\| \geq N p^{-1} L^7 \right) \leq e^{-cL^3}, \quad L = \log(1/p).$$

This is the point where Corollary 6.2 is used: it is applied to the entire path, not separately to each deterministic increment.

After rescaling, the coupling error is bounded by

$$\sqrt{\frac{p}{T(p)}} N p^{-1} L^7 = \frac{N L^7}{\sqrt{p T(p)}}.$$

Since k is fixed and

$$n_i \asymp (m_i p)^{1/3} \asymp (p T(p))^{1/3},$$

we have

$$N = \sum_{i=1}^k (n_i + 1) = O((p T(p))^{1/3}).$$

Hence

$$\frac{N L^7}{\sqrt{p T(p)}} = O \left(\frac{L^7}{(p T(p))^{1/6}} \right) \rightarrow 0.$$

Define the concatenated rescaled increments by

$$\Delta_{p,c}^{(i)} := \sqrt{\frac{p}{T(p)}} (X_c(t_i) - X_c(t_{i-1})).$$

By construction,

$$\Delta_{p,c}^{(i)} = \sqrt{\frac{p}{T(p)}} X^{*,i}(m_i).$$

Moreover, on the good coupling event,

$$\begin{aligned} \|\Delta_p^{(i)} - \Delta_{p,c}^{(i)}\| &\leq \sqrt{\frac{p}{T(p)}} (\|X_{\text{reg}}(t_i) - X_c(t_i)\| + \|X_{\text{reg}}(t_{i-1}) - X_c(t_{i-1})\|) \\ &\leq 2\sqrt{\frac{p}{T(p)}} Np^{-1}L^7. \end{aligned}$$

Therefore, for every fixed i ,

$$\Delta_p^{(i)} - \Delta_{p,c}^{(i)} \xrightarrow{\mathbb{P}} 0.$$

Finally, the variables

$$\Delta_{p,c}^{(1)}, \dots, \Delta_{p,c}^{(k)}$$

are mutually independent, because they depend on disjoint groups of the independent regenerated blocks $X_j^{*,i}$.

Step 3: Gaussian limit for one coupled increment.

Fix i . Let $\sigma_{t'_i}^2$ denote the scalar variance parameter of one regular block at time t'_i , so that by symmetry the covariance matrix of $X_j^{*,i}(t'_i)$ is $\sigma_{t'_i}^2 I_d$. Define

$$Z_j^{(i)} := \frac{X_j^{*,i}(t'_i)}{\sigma_{t'_i}}, \quad j = 1, \dots, n_i.$$

Then, for each fixed i , the variables $Z_j^{(i)}$ are i.i.d. random vectors in $\mathbb{R}^d$.

They are centered by symmetry. Indeed, each regenerated walk starts from the origin with a uniform random initial direction, and the law of the environment is invariant under the symmetries of the lattice. In particular, the law of $X_j^{*,i}(t'_i)$ is symmetric under $x \mapsto -x$, hence

$$\mathbb{E}_p[X_j^{*,i}(t'_i)] = 0, \quad \text{so} \quad \mathbb{E}_p[Z_j^{(i)}] = 0.$$

By definition of $\sigma_{t'_i}^2$, they are normalized in the sense that

$$\text{Cov}(Z_j^{(i)}) = I_d.$$

Moreover, using $H2$ and the same good-event/bad-event decomposition used in the tightness proof, we obtain the fourth-moment estimate

$$\mathbb{E}_p \left[\|X_1^{*,i}(t'_i)\|^4 \right] \leq CL^{32} \frac{(t'_i)^2}{p^2} + \varepsilon_p, \quad L = \log(1/p),$$

where the error term is negligible after the normalizations below. Since $\sigma_{t'_i}^2$ is of order t'_i/p , it follows that

$$\mathbb{E}_p \left[\|Z_1^{(i)}\|^4 \right] \leq CL^{32} + o(1).$$

Thus the fourth moment of the normalized block is not uniformly bounded, but it grows only like a power of $\log(1/p)$.

We now verify Lindeberg's condition for the triangular array

$$\left\{ \frac{1}{\sqrt{n_i}} Z_j^{(i)} \right\}_{1 \leq j \leq n_i}.$$

For every $\eta > 0$, by Markov's inequality,

$$\begin{aligned} \frac{1}{n_i} \sum_{j=1}^{n_i} \mathbb{E}_p \left[\left\| Z_j^{(i)} \right\|^2 \mathbf{1}_{\{\|Z_j^{(i)}\| > \eta \sqrt{n_i}\}} \right] &\leq \frac{1}{\eta^2 n_i^2} \sum_{j=1}^{n_i} \mathbb{E}_p \left\| Z_j^{(i)} \right\|^4 \\ &= \frac{1}{\eta^2 n_i} \mathbb{E}_p \left\| Z_1^{(i)} \right\|^4 \\ &\leq \frac{CL^{32}}{\eta^2 n_i} + o(1). \end{aligned}$$

By our choice $n_i = \lfloor (m_i p)^{1/3} \rfloor$, and since $m_i \sim (\theta_i - \theta_{i-1})T(p)$, we have

$$n_i \asymp (pT(p))^{1/3}.$$

In the polynomial case $T(p) = p^{-m}$ with $m > 11/8$, this diverges polynomially in p^{-1} . Hence

$$\frac{L^{32}}{n_i} \rightarrow 0.$$

Therefore Lindeberg's condition holds. We may therefore apply the multidimensional CLT, which yields

$$\frac{1}{\sqrt{n_i}} \sum_{j=1}^{n_i} Z_j^{(i)} \xrightarrow{d} \mathcal{N}(0, I_d) \quad \text{as } p \rightarrow 0.$$

Now

$$X^{*,i}(m_i) = \sigma_{t'_i} \sum_{j=1}^{n_i} Z_j^{(i)} + X_{n_i+1}^{*,i}(r_i),$$

so

$$\Delta_{p,c}^{(i)} = \sqrt{\frac{p}{T(p)}} \sigma_{t'_i} \sum_{j=1}^{n_i} Z_j^{(i)} + \sqrt{\frac{p}{T(p)}} X_{n_i+1}^{*,i}(r_i).$$

Rewrite the main term as

$$\sqrt{\frac{n_i p \sigma_{t'_i}^2}{T(p)}} \frac{1}{\sqrt{n_i}} \sum_{j=1}^{n_i} Z_j^{(i)}.$$

Since $n_i t'_i = m_i - r_i$, we have

$$\frac{n_i t'_i}{T(p)} \rightarrow \theta_i - \theta_{i-1}.$$

Moreover, by the diffusivity result proved in [1], the variance of the regenerated walk is asymptotically linear in time:

$$\frac{p \sigma_t^2}{t} \rightarrow \sigma^2 \quad \text{as } p \rightarrow 0,$$

uniformly along time scales t in the diffusive regime considered in the paper. Since $t'_i \gg p^{-5/4}$ and $t'_i \leq \bar{T}$, this applies to t'_i , so

$$\frac{p \sigma_{t'_i}^2}{t'_i} \longrightarrow \sigma^2.$$

Therefore

$$\frac{n_i p \sigma_{t'_i}^2}{T(p)} = \frac{n_i t'_i}{T(p)} \frac{p \sigma_{t'_i}^2}{t'_i} \longrightarrow (\theta_i - \theta_{i-1}) \sigma^2.$$

For the remainder term, since $r_i < t'_i$, $H2$ implies that

$$\sqrt{\frac{p}{T(p)}} X_{n_i+1}^{*,i}(r_i) \xrightarrow{\mathbb{P}} 0.$$

Hence, by Slutsky's theorem,

$$\Delta_{p,c}^{(i)} \xrightarrow{d} \mathcal{N}(0, (\theta_i - \theta_{i-1}) \sigma^2 I_d).$$

Step 4: Characteristic functions of the increment vector.

Set

$$D_p := (\Delta_p^{(1)}, \dots, \Delta_p^{(k)}), \quad D_{p,c} := (\Delta_{p,c}^{(1)}, \dots, \Delta_{p,c}^{(k)}).$$

Because the components of $D_{p,c}$ are independent,

$$\varphi_{p,c}(u_1, \dots, u_k) = \prod_{i=1}^k \mathbb{E}_p \left[e^{i \langle u_i, \Delta_{p,c}^{(i)} \rangle} \right].$$

By Step 3, for each fixed i ,

$$\mathbb{E}_p \left[e^{i \langle u_i, \Delta_{p,c}^{(i)} \rangle} \right] \longrightarrow \exp \left(-\frac{\sigma^2}{2} (\theta_i - \theta_{i-1}) \|u_i\|^2 \right).$$

Hence

$$\varphi_{p,c}(u_1, \dots, u_k) \longrightarrow \prod_{i=1}^k \exp \left(-\frac{\sigma^2}{2} (\theta_i - \theta_{i-1}) \|u_i\|^2 \right),$$

which is the characteristic function of a vector of independent Gaussian increments.

Now write

$$\varphi_p(u) - \varphi_{p,c}(u) = \mathbb{E} \left[e^{i \sum_{i=1}^k \langle u_i, \Delta_p^{(i)} \rangle} \left(e^{i \sum_{i=1}^k \langle u_i, \Delta_p^{(i)} - \Delta_{p,c}^{(i)} \rangle} - 1 \right) \right].$$

Therefore

$$|\varphi_p(u) - \varphi_{p,c}(u)| \leq \mathbb{E} \left[\left| e^{i \sum_{i=1}^k \langle u_i, \Delta_p^{(i)} - \Delta_{p,c}^{(i)} \rangle} - 1 \right| \right].$$

Since the integrand is bounded by 2 and converges to 0 in probability, dominated convergence yields

$$|\varphi_p(u) - \varphi_{p,c}(u)| \longrightarrow 0.$$

It follows that

$$\varphi_p(u_1, \dots, u_k) \longrightarrow \prod_{i=1}^k \exp \left(-\frac{\sigma^2}{2} (\theta_i - \theta_{i-1}) \|u_i\|^2 \right).$$

By Lévy's theorem,

$$D_p \xrightarrow{d} (\xi_1, \dots, \xi_k),$$

where the ξ_i are independent and

$$\xi_i \sim \mathcal{N}(0, (\theta_i - \theta_{i-1})\sigma^2 I_d).$$

Step 5: Go back from increments to positions.

Consider the continuous map

$$S : (\mathbb{R}^d)^k \rightarrow (\mathbb{R}^d)^k, \quad S(D_1, \dots, D_k) = (D_1, D_1 + D_2, \dots, D_1 + \dots + D_k).$$

By definition,

$$S(D_p) = (\tilde{Z}_{p,\text{reg}}(\theta_1), \dots, \tilde{Z}_{p,\text{reg}}(\theta_k)),$$

while

$$S(\xi_1, \dots, \xi_k) \stackrel{d}{=} (B(\theta_1), \dots, B(\theta_k)),$$

where B is the Brownian motion with covariance $\text{Cov}(B(t)) = t\sigma^2 I_d$. The continuous mapping theorem gives

$$(\tilde{Z}_{p,\text{reg}}(\theta_1), \dots, \tilde{Z}_{p,\text{reg}}(\theta_k)) \xrightarrow{d} (B(\theta_1), \dots, B(\theta_k)).$$

Finally, by Lemma 5.2, the same convergence holds with $\tilde{Z}_{p,\text{reg}}$ replaced by $Z_{p,\text{reg}}$. The same finite-dimensional convergence then holds for $Z_{p,L}$ by the reduction to the original walk. $\square$

Appendix: Reference Theorems

This appendix collects the classical results invoked in the proof, each stated in the form needed and with the precise reference. For each theorem, we indicate the step and section where it is used.

A.1 Prokhorov's Theorem

Definition A.1 (Polish space). *A metric space $(\mathcal{X}, d)$ is called a Polish space if it is separable and completely metrizable, that is, if there exists a metric inducing the same topology as d under which $\mathcal{X}$ is complete, and $\mathcal{X}$ contains a countable dense subset.*

Definition A.2 (Tightness). *A family $(\mathbb{P}_p)_{p \in (0,1)}$ of probability measures on a Polish space $(\mathcal{X}, d)$ is called tight if for every $\varepsilon > 0$ there exists a compact set $K_\varepsilon \subset \mathcal{X}$ such that*

$$\inf_{p \in (0,1)} \mathbb{P}_p(K_\varepsilon) \geq 1 - \varepsilon.$$

Theorem A.3 (Prokhorov, [2], Theorems 5.1 and 5.2). *Let $(\mathcal{X}, d)$ be a Polish space. A family $(\mathbb{P}_p)$ of probability measures on $\mathcal{X}$ is relatively compact in the weak topology if and only if it is tight.*

A.3 Markov's Inequality

Theorem A.4 (Markov's Inequality). *Let $Z \geq 0$ be a non-negative random variable with $\mathbb{E}[Z] < +\infty$. Then for every $R > 0$:*

$$\mathbb{P}(Z > R) \leq \frac{\mathbb{E}[Z]}{R}.$$

A.4 Arzelà–Ascoli Theorem

Theorem A.5 (Arzelà–Ascoli, [4], Theorem 7.25). *Let $I \subset \mathbb{R}$ be a compact interval. A family $\mathcal{F} \subset C(I, \mathbb{R}^d)$ is relatively compact in the uniform topology if and only if it is:*

- (i) *uniformly bounded;*
- (ii) *uniformly equicontinuous.*

A.5 Multidimensional Central Limit Theorem

Theorem A.6 (Multidimensional Lindeberg–Feller CLT). *Let $\{Z_{n,j}\}_{1 \leq j \leq k_n}$ be a triangular array of centered random vectors in $\mathbb{R}^d$ such that the covariance matrices converge and the Lindeberg condition holds. Then*

$$\sum_{j=1}^{k_n} Z_{n,j} \xrightarrow{d} \mathcal{N}(0, \Sigma).$$

A.6 Layer-Cake Formula

Lemma A.7 (Layer-Cake / Cavalieri's Formula). *Let $Z \geq 0$ be a non-negative random variable. For every $q > 0$:*

$$\mathbb{E}[Z^q] = q \int_0^\infty r^{q-1} \mathbb{P}(Z > r) dr.$$

A.7 L^4 maximal estimate

Lemma A.8 (L^4 maximal estimate). *Let $Y_1, \dots, Y_k$ be independent centered random vectors in $\mathbb{R}^d$, and set*

$$M_\ell := \sum_{j=1}^{\ell} Y_j, \quad 0 \leq \ell \leq k.$$

Then

$$\mathbb{E} \left[\max_{0 \leq \ell \leq k} \|M_\ell\|^4 \right] \leq C \mathbb{E} \|M_k\|^4.$$

Proof. The process $(M_\ell)_{\ell=0}^k$ is a martingale with respect to its natural filtration. Since $x \mapsto \|x\|$ is convex, the process $(\|M_\ell\|)_{\ell=0}^k$ is a non-negative submartingale. Therefore, by Doob's L^4 maximal inequality,

$$\mathbb{E} \left[\max_{0 \leq \ell \leq k} \|M_\ell\|^4 \right] \leq \left(\frac{4}{3} \right)^4 \mathbb{E} \|M_k\|^4.$$

This gives the claim. □

A.8 Fourth moment of block sums

Lemma A.9 (Fourth moment of block sums). *Let*

$$\xi_j := \sqrt{\frac{p}{T(p)}} \Delta_j.$$

There exists a deterministic sequence $\varepsilon_p \rightarrow 0$, with $n_p \varepsilon_p \rightarrow 0$, such that for every consecutive collection I of k block increments,

$$\mathbb{E} \left\| \sum_{j \in I} \xi_j \right\|^4 \leq C \left[\left(\frac{k}{n_p} \right)^2 + \frac{kL^{32}}{n_p^2} + k\varepsilon_p \right].$$

Proof. The variables ξ_j are independent and centered. The centering follows from the uniform initial velocity and symmetry.

First, Lemma 7.1 of [1] gives the correct second-moment scale. Since $s_j - s_{j-1} \in \{r_p, r_p + 1\}$,

$$\mathbb{E} \|\Delta_j\|^2 \leq C \frac{r_p}{p}.$$

Therefore

$$\mathbb{E} \|\xi_j\|^2 = \frac{p}{T(p)} \mathbb{E} \|\Delta_j\|^2 \leq C \frac{r_p}{T(p)} \leq \frac{C}{n_p}.$$

We also need a rough fourth-moment bound for one block. Define

$$A_j := \left\{ \|\Delta_j\| \leq CL^8 \sqrt{\frac{r_p}{p}} \right\}.$$

By $H2$, $\mathbb{P}(A_j^c) \leq Ce^{-2L^2}$. We split

$$\mathbb{E} \|\xi_j\|^4 = \mathbb{E} [\|\xi_j\|^4 \mathbf{1}_{A_j}] + \mathbb{E} [\|\xi_j\|^4 \mathbf{1}_{A_j^c}].$$

On A_j ,

$$\|\xi_j\| \leq CL^8 \sqrt{\frac{r_p}{T(p)}} \leq \frac{CL^8}{\sqrt{n_p}},$$

so

$$\mathbb{E} [\|\xi_j\|^4 \mathbf{1}_{A_j}] \leq C \frac{L^{32}}{n_p^2}.$$

On A_j^c , we use the deterministic nearest-neighbor bound $\|\Delta_j\| \leq 2r_p$. Hence

$$\mathbb{E} [\|\xi_j\|^4 \mathbf{1}_{A_j^c}] \leq C \left(\sqrt{\frac{p}{T(p)}} r_p \right)^4 e^{-2L^2}.$$

The polynomial factor is killed by e^{-2L^2} . Therefore this bad-event term is bounded by a sequence $\varepsilon_p \rightarrow 0$, and we may choose it so that $n_p \varepsilon_p \rightarrow 0$, since n_p is polynomial in p^{-1} . Thus

$$\mathbb{E} \|\xi_j\|^4 \leq C \frac{L^{32}}{n_p^2} + \varepsilon_p.$$

Now let $S_I := \sum_{j \in I} \xi_j$. For a fixed coordinate a , we use the standard fourth-moment expansion for independent centered variables. Indeed, after expanding the fourth

power, every mixed term with an index appearing only once has expectation zero. Thus

$$\mathbb{E}|S_{I,a}|^4 = \sum_{j \in I} \mathbb{E}|\xi_{j,a}|^4 + 6 \sum_{\substack{i,j \in I \\ i < j}} \mathbb{E}|\xi_{i,a}|^2 \mathbb{E}|\xi_{j,a}|^2,$$

and therefore

$$\mathbb{E}|S_{I,a}|^4 \leq C \left[\left(\sum_{j \in I} \mathbb{E}|\xi_{j,a}|^2 \right)^2 + \sum_{j \in I} \mathbb{E}|\xi_{j,a}|^4 \right].$$

Using the bounds above,

$$\sum_{j \in I} \mathbb{E}|\xi_{j,a}|^2 \leq C \frac{k}{n_p}, \quad \sum_{j \in I} \mathbb{E}|\xi_{j,a}|^4 \leq C \frac{kL^{32}}{n_p^2} + k\varepsilon_p.$$

Since d is fixed, summing over coordinates gives the claim. $\square$

Theorem A.10 (Billingsley's tightness criterion in $C([0, \Theta], \mathbb{R}^d)$) *Let $\{P_n\}_{n \geq 1}$ be a sequence of probability measures on $C([0, \Theta], \mathbb{R}^d)$, endowed with the uniform topology. For $f \in C([0, \Theta], \mathbb{R}^d)$, define its modulus of continuity by*

$$\omega_f(\delta) := \sup_{\substack{0 \leq s, t \leq \Theta \\ |t-s| \leq \delta}} \|f(t) - f(s)\|.$$

Then $\{P_n\}_{n \geq 1}$ is tight if and only if the following two conditions hold:

1. *for every $\eta > 0$, there exist $a > 0$ and $n_0 \geq 1$ such that*

$$P_n(\{f : \|f(0)\| \geq a\}) \leq \eta, \quad n \geq n_0;$$

2. *for every $\varepsilon > 0$,*

$$\lim_{\delta \downarrow 0} \limsup_{n \rightarrow \infty} P_n(\{f : \omega_f(\delta) \geq \varepsilon\}) = 0.$$

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